

Nathan Critchett

AI STRATEGIST · BUILDER · AUTHOR

I close the gap between what AI can produce and what the people using it can evaluate.
Strategy, shipped tools, and adoption, grounded in cognitive science.

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PROFILE

AI strategist and hands-on builder working at the intersection of AI capability and human capability. I translate cognitive-science research into **shipped products, LLM-assisted tools, and adoption programs**, drawing on field work across **100+ organizations**. Two live platforms, an interactive diagnostic engine, a forthcoming book, and **60+ published articles** on AI strategy, learning design, and the future of work.

CORE COMPETENCIES

AI Strategy

Generative AI

Large Language Models (LLMs)

Prompt Engineering

AI Adoption & Change

Human-in-the-Loop Design

Rapid Prototyping

Product Strategy

Learning Experience Design

Instructional Design

EdTech

Assessment & Diagnostics

Computer Vision

JavaScript

Python

Serverless (Cloudflare Workers)

Technical Writing

Keynote Speaking

Stakeholder Engagement

SELECTED TOOLS, PRODUCTS & PROTOTYPES

Arq.Training / Founder & Builder

● Live · arq.training

Cognitive diagnostic platform for education. Maps how students reason, not just what they score. Built on the Model of Hierarchical Complexity.

LLM-assisted analysis · assessment design · EdTech

CritchPitch / Founder & Builder

● Live · critchpitch.com

Screens youth pitching mechanics and injury risk from a single phone video. Paired with a 25-article, fully cited arm-health library.

Computer vision · mobile capture · content engine

The Cognitive Audit / Designer & Developer

● Live · /audit

Interactive 15-question diagnostic. A weighted scoring engine surfaces the weakest framework and the single first action to take.

JavaScript · scoring engine · lead capture

AI-Native Portfolio CRM / Prototype

In development

Context-extraction system that surfaces cross-portfolio patterns while keeping a human at the center of every decision.

LLM · retrieval · human-in-the-loop

Architects List / Serverless backend

Shipped

Automated signup and welcome-kit delivery powering the book and audit funnels.

Cloudflare Workers · KV storage · Resend API · auth dashboard

Solo-built publishing system. A programmatic pipeline generates Open Graph cards, sitemaps, and schema.org structured data across three libraries.

Python · Node · structured data · analytics

AUTHOR & PUBLISHED RESEARCH

Cognitive Architecture: How to Think When Machines Think For You. Author, 2026. A field guide for leaders built from data inside 100+ organizations. Five frameworks: the Cognitive Supply Chain, the Centaur Architecture, the Antifragile Ego, the Horizontal/Vertical distinction, and the shift from worker to Architect of Intelligence.

8 whitepapers and 16 essays on AI and cognition. Including The Complexity Gap, Cognitive Offloading, Universal Basic Upgrading, and Stop Grading AI Output. Grounded in cognitive science and tested in the field.

60+ evidence-based articles across three libraries. AI and Cognition (24), Cognitive Development (14), Youth Pitching (25). Each fully sourced.

SPEAKING & ADVISORY

Keynotes, half- and full-day sessions, and professional development on AI strategy and cognitive development. Signature talks: **The Complexity Gap**, **AI as Mirror, Not GPS**, and **Building the Thinkers the Future Needs**. District work delivered with **Edapt** and Christian Jackson. Selected audiences:

- ◆ **Greater Ozarks Cooperating School Districts** (GOCSO)
- ◆ **AAEA**, educational administrators
- ◆ **LA County Fire Department**, Lifeguard Division

- ◆ **Association of California School Administrators** (ACSA)
- ◆ **Small School Districts' Association** (SSDA)
- ◆ **Multiple K-12 school districts**

VENTURES & EARLIER

Arq.Training, Founder. Cognitive diagnostics for education.

CritchPitch, Founder. Youth pitching screening and arm-health.

Division I pitcher and sports-medicine All-American. Former first responder, LA County Fire Department, Lifeguard Division (ocean rescue).